ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

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Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer: 1. There are 594 documents in the category sci.med. 2. By pre-splitting each field and calculating the number of entries for that field, the algorithm avoids spending time on calculations, improving efficiency. Furthermore, the selected categories can be dynamically adjusted later, avoiding the need to recalculate everything every time a change is made.

AI used for this answer: No

If yes, how:

External references (links):

Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer: The presence of such a large amount of data with errors in a particular field may indicate past errors in field selection or data recording. However, this does not mean the data is entirely worthless. Directly deleting the data may lead to biased content, resulting in incomplete and poorly interpretable data analysis results. Therefore, other methods can be used, such as estimating possible values. In this case, the category name is derived from the category number, so if the category still exists, the correct content can be deduced.

AI used for this answer: No

If yes, how:

External references (links):

Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer: 1. No the sampling result cannot represent the population result. For example, the ratio of the population of comp.graphics is 599/2257=0.265. However the ratio of the sampling is 278/1000=0.278. The two ratio is close but a sight gap still remains. 2. Because sampling cannot perfectly represent the population, sampling on a extremely imbalanced dataset must cause basis. For example, making a imbalanced dataset looks balance after sampling.

AI used for this answer: No

If yes, how:

External references (links):

Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

Questions:

1. Does the total number of features (columns) increase or decrease?
2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer: 1. Increase. The shape is 35482 when using 'english' as a stop word. After changing the stop word to None, which does no filter on words, the shape increase to 35788. 2. Sparsity will be lower. Common stop words like 'the' or 'and' can be in every article. If adding those words back, the ratio of zero will decrease, leading to lower the sparsity.

AI used for this answer: No

If yes, how:

External references (links):

Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer: 1. increasing in threshold makes kept dramatically decrease. Below table is the amount of keep==True under different threshold. 2. Extremely rare words will be eliminated first because these words only show in very few documents, the variances are nearly zero. In addition, if not filter the stop words in advanced, those common words showing in every document will also be eliminated due to lack of diversity. 3. It can preserve the differences between documents while reducing noise, text with low variability provides very little information but occupies dimensional space, which is interference noise for analysis.

Threshold	Number of terms kept (keep == True)
0.0	35482
0.01	5089
0.05	1491

AI used for this answer: No

If yes, how:

External references (links):

Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
2. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?
3. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

Answer: 1. Based on my understanding of computer graphics software and hardware, Pearson's top 10 has more words that should be classified as stop words, with only three professional words "vga", "video", and "card"; Spearman has more than half of the professional words such as “3d”, “graphics”, "files", "image", "program", "file", "windows". 2. The former only assigns a value of 1 to linear relationships, while the latter measures monotonic relationships. In other words, correlation between two variables usually results in a higher evaluation. 3. Data that is correlated but non-linear, such as "when the temperature is 10, it sells 10 cups of coffee," or "at 20 degrees Celsius, it sells 20 cups," or "at 30 degrees Celsius, it sells 23 cups," can also occur when a feature (word frequency) has extreme outliers. For example, a word might appear 1 to 2 times in most articles, but an anomalously high number of times (500 times) might appear 500 times in one article. This extreme value would severely strain Pearson's linear fit, causing its score to plummet; however, Spearman only considers the "ranking" of the numerical value (500 times is ranked greater than 2), and is unaffected by the distance between extreme values, thus maintaining a high relevance score.

AI used for this answer: No

If yes, how:

External references (links):

Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup:

Both the number of patterns identified and the number of runtimes increased significantly. Setting a lower frequency resulted in more words passing through the criteria, allowing the model to capture more patterns, thus increasing the runtime.

minsup	#pattern	Runtime
3	9547	2.644
9	710	0.044

Answer (2) — Top-K k vs MaxFPGrowth minSup:

minSup is an "absolute threshold" that will find any pattern greater than the set number. It is a bit difficult to set when you are not familiar with the data. If it is too large, it will not be found; if it is too small, too many patterns will be found. topk is a "relative ranking" that will always return the top k data points. That is when k =100, it lists 100 rows. It can find a very stable number of patterns.

Answer (3) — Maximal vs standard frequent patterns:

Because the Maximal pattern only retains the longest combinations that are not included in longer frequent patterns. Ordinary frequent patterns output all possible subsets (e.g., if there is {A, B}, it will also output {A} and {B}), generating a lot of redundant information; while the Maximal pattern filters out these redundant subsets, thus producing a more concise result in structure, effectively reducing the number of features and redundancy.

AI used for this answer: No

If yes, how:

External references (links):

Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer: 1. Each feature (a single word in the matrix) represents a "dimensionality." The X_counts matrix contains tens of thousands of words, meaning this data resides in a mathematical space with "tens of thousands of dimensions." Human vision and standard scatter plots can only display 2-dimensional (X and Y axes) or 3-dimensional (X, Y, and Z axes) images at most. We cannot draw and understand charts with tens of thousands of coordinate axes on a screen; therefore, direct visualization is physically and mathematically impossible. 2. Dimensionality reduction techniques such as PCA, t-SNE, and UMAP are applied. The core function of these algorithms is to condense and compress tens of thousands of dimensions of data into only 2 or 3 dimensions through mathematical operations, while preserving as much of the original relationships and structure between articles as possible. After dimensionality reduction, you can then place them on a 2D scatter plot to see if these four news groups cluster into clear blocks.

AI used for this answer: No

If yes, how:

External references (links):

Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer: 1. Both t-SNE and UMAP show a tendency towards category partitioning (blue and yellow each occupy one side), but they don't quite reach the point where clean cluster islands can be distinguished. However, there is still significant mixing in the center, especially in sci.med and comp.graphics. PCA shows the most severe overlap, with the four categories highly concentrated and overlapping in the middle of the far left. 2. PCA is a linear projection where the new axes are linear combinations of the original features. PCA seeks directions that are orthogonal to each other and maximize the variance after projection, making it well-suited for highly structured data. High-dimensional text data contains complex, non-linear relationships that a linear projection cannot capture, whereas t-SNE and UMAP are non-linear techniques explicitly designed to unravel and preserve these intricate local structures. 3. t-sne calculates the probability n_ij of a point being a neighbor. Close proximity indicates a high likelihood of a neighbor, suggesting high similarity. However, this only indicates local similarity and cannot be inferred from overall similarity.

AI used for this answer: Yes

If yes, how: l don’t understand pca, t-SNE, and UMAP for a long time after I taking the class and read the description in master.ipynb. so I ask AI to guide me what they do and how to interpret their results.

External references (links): https://leemeng.tw/essence-of-principal-component-analysis.html

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Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?
3. How does one-hot binarization eliminate this false assumption?

Answer: 1. four. But it can actually use only three. 2. It may lead the model to interpret the number, not categories. For example, KNN could do calculate in the number and interpret the difference as the distance between the categories. Also, the model will offer different results after reorganizing the number. 3. Each category has its own column, using only 0s and 1s for comparison. Any two categories are equidistant, preventing misinterpretation of the numbers.

AI used for this answer: No

If yes, how:

External references (links):
